PHYDSC403T — Basic Quantum Mechanics

Unit 4: Hydrogen Atom and Linear Harmonic Oscillator

7th Semester, Assam University (FYUG) — Haflong Government College

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Notes 14 of 20 — drafted by gemini-3.6-flash (Kaggle Model Proxy) from the unit syllabus only, without external sources, and checked against the syllabus.

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Unit syllabus (as in the syllabus file)

1. Wave equation for motion of electron, ground state of hydrogen atom, most probable position of electron in the ground state.
2. Application of Schrodinger Equation to find energy eigen value and eigen function of hydrogen atom using spherical harmonics and Laguerre polynomials.
3. Degeneracy of energy levels of hydrogen atom.
4. Linear harmonic oscillator: Energy eigen value and eigen function of linear harmonic oscillator, ground state wave function.

Top 2 high-mark topics in this unit

1. Hydrogen atom: Schrodinger equation in spherical polar coordinates, radial wave equation, energy eigenvalues and eigenfunctions using spherical harmonics and Laguerre polynomials, and most probable distance — 3 of 5 AU questions ask this separation plus most probable distance, repeated in 2020, 2021 and 2023.
2. Linear harmonic oscillator: energy eigenvalues and eigenfunctions, ground-state wave function, and zero-point energy — AU links it to the H-atom question in 2021 and to uncertainty / zero-point energy in U1-U2, and it is a standard derivation.

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1. Wave Equation for the Motion of the Electron in the Hydrogen Atom

1.1 Time-Independent Schrödinger Equation in Cartesian Coordinates

The hydrogen atom consists of a positively charged nucleus of charge $+e$ fixed at the origin and an electron of mass m and charge $-e$ moving around it under the influence of an attractive electrostatic Coulomb potential. The electrostatic potential energy $V(r)$ of the electron at a distance r from the nucleus in SI units is given by

$$V(r) = -\frac{e^2}{4\pi\varepsilon_0 r}$$

where ε_0 is the permittivity of free space. In Cartesian coordinates (x, y, z), the distance r from the origin is expressed as

$$r = \sqrt{x^2 + y^2 + z^2}$$

Substituting this potential energy into the three-dimensional time-independent Schrödinger equation yields

$$-\frac{\hbar^2}{2m}\nabla^2\psi(x,y,z) + V(r)\psi(x,y,z) = E\psi(x,y,z)$$

where $\hbar = \frac{h}{2\pi}$ is the reduced Planck constant, E is the total energy of the electron, and ∇^2 is the Laplacian operator. Expanding the Laplacian operator explicitly in Cartesian coordinates gives

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

Inserting the expression for ∇^2 and the Coulomb potential $V(r)$ into the Schrödinger equation yields

$$-\frac{\hbar^2}{2m}\left(\frac{\partial^2\psi}{\partial x^2} + \frac{\partial^2\psi}{\partial y^2} + \frac{\partial^2\psi}{\partial z^2}\right) - \frac{e^2}{4\pi\varepsilon_0\sqrt{x^2+y^2+z^2}}\psi = E\psi$$

1.2 Transformation to Spherical Polar Coordinates

Because the Coulomb potential $V(r)$ possesses spherical symmetry and depends solely on the radial distance r, it is advantageous to transform the wave equation from Cartesian coordinates (x, y, z) to spherical polar coordinates (r, θ, ϕ). The transformation relations between Cartesian and spherical polar coordinates are defined by

$$x = r\sin\theta\cos\phi$$

$$y = r\sin\theta\sin\phi$$

$$z = r\cos\theta$$

where $r \geq 0$, $0 \leq \theta \leq \pi$, and $0 \leq \phi < 2\pi$.

The Laplacian operator ∇^2 in spherical polar coordinates is expressed as

$$\nabla^2 = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2}{\partial\phi^2}$$

Substituting this representation of ∇^2 into the time-independent Schrödinger equation leads to

$$\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial\psi}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial\psi}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2\psi}{\partial\phi^2} + \frac{2m}{\hbar^2}\left(E + \frac{e^2}{4\pi\varepsilon_0 r}\right)\psi = 0$$

1.3 Separation of Variables

To solve this partial differential equation, the technique of separation of variables is applied. We assume that the total wave function $\psi(r,\theta,\phi)$ can be expressed as a product of a radial function $R(r)$ depending only on r and an angular function $Y(\theta,\phi)$ depending on θ and ϕ. Furthermore, $Y(\theta,\phi)$ is factored into functions of θ and ϕ individually:

$$\psi(r,\theta,\phi) = R(r)Y(\theta,\phi) = R(r)\Theta(\theta)\Phi(\phi)$$

Substituting this product form into the transformed Schrödinger equation gives

$$\frac{\Theta\Phi}{r^2}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right) + \frac{R\Phi}{r^2\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right) + \frac{R\Theta}{r^2\sin^2\theta}\frac{d^2\Phi}{d\phi^2} + \frac{2m}{\hbar^2}\left(E + \frac{e^2}{4\pi\varepsilon_0 r}\right)R\Theta\Phi = 0$$

Dividing the entire equation by the product $R(r)\Theta(\theta)\Phi(\phi)$ separates the terms:

$$\frac{1}{r^2R}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right) + \frac{1}{r^2\Theta\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right) + \frac{1}{r^2\Phi\sin^2\theta}\frac{d^2\Phi}{d\phi^2} + \frac{2m}{\hbar^2}\left(E + \frac{e^2}{4\pi\varepsilon_0 r}\right) = 0$$

Multiplying the entire equation by $r^2\sin^2\theta$ isolates the ϕ-dependent term on one side:

$$\frac{\sin^2\theta}{R}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right) + \frac{\sin\theta}{\Theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right) + \frac{2mr^2\sin^2\theta}{\hbar^2}\left(E + \frac{e^2}{4\pi\varepsilon_0 r}\right) = -\frac{1}{\Phi}\frac{d^2\Phi}{d\phi^2}$$

The left-hand side of this equation depends only on r and θ, while the right-hand side depends only on ϕ. For this identity to hold for all values of the independent variables, both sides must equal a common separation constant, which is conventionally designated as m_l^2.

1.4 The Azimuthal (ϕ) Equation

Equating the right-hand side of the separated equation to m_l^2 yields the differential equation for $\Phi(\phi)$:

$$-\frac{1}{\Phi}\frac{d^2\Phi}{d\phi^2} = m_l^2$$

Rearranging this gives the standard harmonic oscillator equation:

$$\frac{d^2\Phi}{d\phi^2} + m_l^2\Phi = 0$$

The general solution to this ordinary differential equation is

$$\Phi(\phi) = Ae^{im_l\phi}$$

where A is a normalization constant.

For the wave function to be single-valued in physical space, the solution must satisfy the periodic boundary condition $\Phi(\phi + 2\pi) = \Phi(\phi)$. Applying this condition:

$$Ae^{im_l(\phi+2\pi)} = Ae^{im_l\phi} \implies e^{im_l 2\pi} = 1$$

This condition requires m_l to be an integer:

$$m_l = 0, \pm 1, \pm 2, \pm 3, \ldots$$

where m_l is called the magnetic quantum number.

The normalization constant A is determined by setting the integral of $|\Phi(\phi)|^2$ over the interval $[0, 2\pi]$ equal to unity:

$$\int_0^{2\pi} |\Phi(\phi)|^2 d\phi = 1$$

Substituting $\Phi(\phi) = Ae^{im_l\phi}$:

$$|A|^2 \int_0^{2\pi} e^{-im_l\phi} e^{im_l\phi} d\phi = 1$$

$$|A|^2 \int_0^{2\pi} d\phi = 1 \implies 2\pi|A|^2 = 1 \implies A = \frac{1}{\sqrt{2\pi}}$$

Thus, the normalized azimuthal wave function is

$$\Phi(\phi) = \frac{1}{\sqrt{2\pi}} e^{im_l\phi}$$

1.5 The Polar (θ) Equation

Equating the left-hand side of the separated equation to m_l^2 gives

$$\frac{\sin^2\theta}{R}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right) + \frac{\sin\theta}{\Theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right) + \frac{2mr^2\sin^2\theta}{\hbar^2}\left(E + \frac{e^2}{4\pi\varepsilon_0 r}\right) = m_l^2$$

Dividing this equation by $\sin^2\theta$ allows complete separation of the radial variable r and polar variable θ:

$$\frac{1}{R}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right) + \frac{2mr^2}{\hbar^2}\left(E + \frac{e^2}{4\pi\varepsilon_0 r}\right) = -\frac{1}{\Theta\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right) + \frac{m_l^2}{\sin^2\theta}$$

Since the left-hand side depends only on r and the right-hand side depends only on θ, both sides must be equal to a second separation constant, designated as $l(l+1)$.

Setting the right-hand side equal to $l(l+1)$ gives the polar differential equation:

$$-\frac{1}{\Theta \sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right)+\frac{m_l^2}{\sin^2\theta}=l(l+1)$$

Rearranging terms yields

$$\frac{1}{\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right)+\left[l(l+1)-\frac{m_l^2}{\sin^2\theta}\right]\Theta(\theta)=0$$

By making the change of variable $x=\cos\theta$, this equation transforms into the Associated Legendre Differential Equation. To obtain physically acceptable solutions that remain finite, continuous, and single-valued throughout the interval $\theta\in[0,\pi]$ (specifically at the poles where $\cos\theta=\pm1$), the separation constant l must be a non-negative integer:

$$l=0,1,2,3,\ldots$$

where l is called the orbital angular momentum quantum number. Furthermore, the solutions exist only when the absolute value of the magnetic quantum number m_l is bounded by l:

$$|m_l|\le l \quad\Longrightarrow\quad m_l=-l,-l+1,\ldots,0,\ldots,l-1,l$$

The normalized solutions $\Theta(\theta)$ are given in terms of the Associated Legendre functions $P_l^{|m_l|}(\cos\theta)$.

1.6 The Radial Equation

Equating the radial part of the separated equation to the constant $l(l+1)$ gives

$$\frac{1}{R}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right)+\frac{2mr^2}{\hbar^2}\left(E+\frac{e^2}{4\pi\varepsilon_0 r}\right)=l(l+1)$$

Dividing by r^2 and rearranging terms yields the radial Schrödinger equation for $R(r)$:

$$\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right)+\left[\frac{2m}{\hbar^2}\left(E+\frac{e^2}{4\pi\varepsilon_0 r}\right)-\frac{l(l+1)}{r^2}\right]R(r)=0$$

To cast this equation into a simplified one-dimensional form, we introduce the radial function transformation

$$u(r)=rR(r)$$

Rearranging this definition expresses $R(r)$ in terms of $u(r)$:

$$R(r)=\frac{u(r)}{r}$$

Taking the first derivative of $R(r)$ with respect to r using the quotient rule:

$$\frac{dR}{dr}=\frac{d}{dr}\left(\frac{u}{r}\right)=\frac{1}{r}\frac{du}{dr}-\frac{u}{r^2}$$

Multiplying by r^2:

$$r^2\frac{dR}{dr} = r\frac{du}{dr} - u$$

Differentiating this product with respect to r:

$$\frac{d}{dr}\left(r^2\frac{dR}{dr}\right) = \frac{d}{dr}\left(r\frac{du}{dr} - u\right) = \frac{du}{dr} + r\frac{d^2u}{dr^2} - \frac{du}{dr} = r\frac{d^2u}{dr^2}$$

Substituting this result back into the radial differential equation gives

$$\frac{1}{r^2}\left(r\frac{d^2u}{dr^2}\right) + \left[\frac{2m}{\hbar^2}\left(E + \frac{e^2}{4\pi\varepsilon_0 r}\right) - \frac{l(l+1)}{r^2}\right]\frac{u}{r} = 0$$

Dividing through by $1/r$ leads directly to the radial wave equation for $u(r)$:

$$\frac{d^2u}{dr^2} + \left[\frac{2m}{\hbar^2}\left(E + \frac{e^2}{4\pi\varepsilon_0 r}\right) - \frac{l(l+1)}{r^2}\right]u(r) = 0$$

2. Energy Eigenvalues and Eigenfunctions of the Hydrogen Atom

Separation of the Time-Independent Schrödinger Equation

The time-independent Schrödinger equation for an electron of mass m moving in the Coulomb potential $V(r) = -\frac{e^2}{4\pi\varepsilon_0 r}$ of a hydrogen nucleus is written as

$$-\frac{\hbar^2}{2m}\nabla^2\psi(r,\theta,\phi) + V(r)\psi(r,\theta,\phi) = E\psi(r,\theta,\phi)$$

In spherical polar coordinates (r, θ, ϕ), the Laplacian operator ∇^2 takes the form

$$\nabla^2 = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2}{\partial\phi^2}$$

Substituting this expression for the Laplacian operator into the Schrödinger equation yields

$$-\frac{\hbar^2}{2m}\left[\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial\psi}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial\psi}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2\psi}{\partial\phi^2}\right] - \frac{e^2}{4\pi\varepsilon_0 r}\psi = E\psi$$

We seek factorized solutions of the wave function by separating the radial and angular variables:

$$\psi(r,\theta,\phi) = R(r)Y(\theta,\phi)$$

Substituting this product form of the wave function into the full Schrödinger equation gives

$$-\frac{\hbar^2}{2m}\left[\frac{Y}{r^2}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right) + \frac{R}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial Y}{\partial\theta}\right) + \frac{R}{r^2\sin^2\theta}\frac{\partial^2 Y}{\partial\phi^2}\right] - \frac{e^2}{4\pi\varepsilon_0 r}RY = ERY$$

Dividing the entire differential equation by $R(r)Y(\theta,\phi)$ and multiplying by $-\frac{2mr^2}{\hbar^2}$, we rearrange terms to isolate the radial dependence on the left-hand side and the angular dependence on the right-hand side:

$$\frac{1}{R}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right)+\frac{2mr^2}{\hbar^2}\left(E+\frac{e^2}{4\pi\varepsilon_0 r}\right)=-\frac{1}{Y}\left[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial Y}{\partial\theta}\right)+\frac{1}{\sin^2\theta}\frac{\partial^2 Y}{\partial\phi^2}\right]$$

Because the left-hand side depends only on r and the right-hand side depends only on θ and ϕ, both sides must be equal to a common separation constant. We set this separation constant equal to $l(l+1)$, where l is a non-negative integer ($l=0,1,2,\ldots$).

The Angular Solutions: Spherical Harmonics

Equating the angular portion of the separated equation to the constant $l(l+1)$ gives

$$-\left[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial Y}{\partial\theta}\right)+\frac{1}{\sin^2\theta}\frac{\partial^2 Y}{\partial\phi^2}\right]=l(l+1)Y(\theta,\phi)$$

The single-valued, continuous, and normalizable solutions to this differential equation are the spherical harmonics $Y_l^{m_l}(\theta,\phi)$, specified by the orbital quantum number l and the magnetic quantum number m_l, where m_l takes $(2l+1)$ integer values ranging from $-l$ to $+l$:

$$Y_l^{m_l}(\theta,\phi)=(-1)^{m_l}\sqrt{\frac{(2l+1)}{4\pi}\frac{(l-m_l)!}{(l+m_l)!}}P_l^{m_l}(\cos\theta)e^{im_l\phi}$$

where $P_l^{m_l}(\cos\theta)$ represents the associated Legendre polynomials.

The Radial Equation and Asymptotic Analysis

Equating the radial portion of the separated equation to $l(l+1)$ yields the radial differential equation:

$$\frac{1}{R}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right)+\frac{2mr^2}{\hbar^2}\left(E+\frac{e^2}{4\pi\varepsilon_0 r}\right)=l(l+1)$$

Multiplying through by $R(r)$ and evaluating the derivative on the left-hand side gives

$$\frac{d^2R}{dr^2}+\frac{2}{r}\frac{dR}{dr}+\left[\frac{2mE}{\hbar^2}+\frac{2me^2}{4\pi\varepsilon_0\hbar^2 r}-\frac{l(l+1)}{r^2}\right]R=0$$

For bound state solutions of the hydrogen atom, the total energy E is negative. We define a positive parameter γ by the relation

$$\gamma^2=-\frac{8mE}{\hbar^2}$$

Substituting γ^2 into the radial equation yields

$$\frac{d^2R}{dr^2}+\frac{2}{r}\frac{dR}{dr}+\left[-\frac{\gamma^2}{4}+\frac{2me^2}{4\pi\varepsilon_0\hbar^2 r}-\frac{l(l+1)}{r^2}\right]R=0$$

To transform the equation into a dimensionless form, we introduce a new variable ρ defined as

$$\rho = \gamma r$$

Using the chain rule, the derivatives with respect to r transform into derivatives with respect to ρ:

$$\frac{dR}{dr} = \frac{d\rho}{dr}\frac{dR}{d\rho} = \gamma\frac{dR}{d\rho}$$

$$\frac{d^2R}{dr^2} = \gamma^2\frac{d^2R}{d\rho^2}$$

Substituting these derivative transformations into the radial equation gives

$$\gamma^2\frac{d^2R}{d\rho^2} + \frac{2\gamma}{\rho}\gamma\frac{dR}{d\rho} + \left[-\frac{\gamma^2}{4} + \frac{2me^2\gamma}{4\pi\varepsilon_0\hbar^2\rho} - \frac{l(l+1)\gamma^2}{\rho^2}\right]R = 0$$

Dividing the entire equation by γ^2 simplifies the expression to

$$\frac{d^2R}{d\rho^2} + \frac{2}{\rho}\frac{dR}{d\rho} + \left[-\frac{1}{4} + \frac{me^2}{4\pi\varepsilon_0\hbar^2\gamma\rho} - \frac{l(l+1)}{\rho^2}\right]R = 0$$

We now define a dimensionless parameter ρ_0 as

$$\rho_0 = \frac{me^2}{4\pi\varepsilon_0\hbar^2\gamma}$$

Rewriting the radial differential equation in terms of ρ_0 gives

$$\frac{d^2R}{d\rho^2} + \frac{2}{\rho}\frac{dR}{d\rho} + \left[\frac{\rho_0}{\rho} - \frac{1}{4} - \frac{l(l+1)}{\rho^2}\right]R = 0$$

To solve this equation, we first determine its asymptotic behavior in the two extreme limits of ρ:

1. In the limit $\rho \to \infty$, the terms proportional to $1/\rho$ and $1/\rho^2$ become negligible, leaving the simplified differential equation

$$\frac{d^2R}{d\rho^2} - \frac{1}{4}R = 0$$

The physically acceptable solution that remains finite and vanishes as $\rho \to \infty$ is

$$R(\rho) \sim e^{-\rho/2}$$

1. In the limit $\rho \to 0$, the centrifugal term $-\frac{l(l+1)}{\rho^2}$ dominates inside the bracket, reducing the differential equation to

$$\frac{d^2R}{d\rho^2} + \frac{2}{\rho}\frac{dR}{d\rho} - \frac{l(l+1)}{\rho^2}R = 0$$

Assuming a trial power solution $R(\rho) \sim \rho^s$ yields the characteristic equation $s(s+1) = l(l+1)$, which gives the roots $s = l$ and $s = -(l+1)$. To ensure that the radial wave function remains finite at the origin $\rho = 0$, we choose the positive root $s = l$, giving

$$R(\rho) \sim \rho^l$$

Combining the asymptotic behaviors at both boundaries, we express the full solution $R(\rho)$ as

$$R(\rho) = e^{-\rho/2}\rho^l v(\rho)$$

Power-Series Solution and Laguerre Polynomials

To derive the differential equation for $v(\rho)$, we first compute the first derivative of $R(\rho)$ with respect to ρ:

$$\frac{dR}{d\rho} = e^{-\rho/2}\rho^l \frac{dv}{d\rho} + le^{-\rho/2}\rho^{l-1}v - \frac{1}{2}e^{-\rho/2}\rho^l v$$

$$\frac{dR}{d\rho} = e^{-\rho/2}\rho^l \left[\frac{dv}{d\rho} + \left(\frac{l}{\rho} - \frac{1}{2}\right)v\right]$$

Differentiating a second time with respect to ρ gives

$$\frac{d^2R}{d\rho^2} = e^{-\rho/2}\rho^l \left[\frac{d^2v}{d\rho^2} + 2\left(\frac{l}{\rho} - \frac{1}{2}\right)\frac{dv}{d\rho} + \left(\frac{l(l-1)}{\rho^2} - \frac{l}{\rho} + \frac{1}{4}\right)v\right]$$

Substituting $R(\rho)$, $\frac{dR}{d\rho}$, and $\frac{d^2R}{d\rho^2}$ back into the radial differential equation and canceling out the common exponential factor $e^{-\rho/2}\rho^l$ leads to the canonical differential equation for $v(\rho)$:

$$\rho\frac{d^2v}{d\rho^2} + [2(l+1) - \rho]\frac{dv}{d\rho} + [\rho_0 - (l+1)]v = 0$$

We solve for $v(\rho)$ using the power-series method by assuming an expansion of the form

$$v(\rho) = \sum_{k=0}^{\infty} c_k \rho^k$$

Differentiating this series term by term yields

$$\frac{dv}{d\rho} = \sum_{k=0}^{\infty} k c_k \rho^{k-1}$$

$$\frac{d^2v}{d\rho^2} = \sum_{k=0}^{\infty} k(k-1) c_k \rho^{k-2}$$

Substituting these series expansions into the differential equation for $v(\rho)$ yields

$$\rho\sum_{k=0}^{\infty} k(k-1)c_k\rho^{k-2} + [2(l+1) - \rho]\sum_{k=0}^{\infty} k c_k \rho^{k-1} + [\rho_0 - l - 1]\sum_{k=0}^{\infty} c_k \rho^k = 0$$

Expanding the terms and grouping by powers of ρ gives

$$\sum_{k=0}^{\infty} k(k-1)c_k\rho^{k-1} + 2(l+1)\sum_{k=0}^{\infty} kc_k\rho^{k-1} - \sum_{k=0}^{\infty} kc_k\rho^k + (\rho_0 - l - 1)\sum_{k=0}^{\infty} c_k\rho^k = 0$$

Shifting the index of the first two summation terms from k to $k+1$ yields

$$\sum_{k=0}^{\infty}(k+1)(k+2l+2)c_{k+1}\rho^k + \sum_{k=0}^{\infty}(\rho_0 - l - 1 - k)c_k\rho^k = 0$$

Combining both summations into a single equation gives

$$\sum_{k=0}^{\infty}\left[(k+1)(k+2l+2)c_{k+1} + (\rho_0 - l - 1 - k)c_k\right]\rho^k = 0$$

For this power series to equal zero identically for all values of ρ, the coefficient of each power ρ^k must equal zero:

$$(k+1)(k+2l+2)c_{k+1} + (\rho_0 - l - 1 - k)c_k = 0$$

Rearranging this condition gives the recurrence relation for the coefficients of the series:

$$c_{k+1} = \frac{k + l + 1 - \rho_0}{(k+1)(k+2l+2)}c_k$$

Quantization of Energy Eigenvalues

For very large values of k, the recurrence relation approaches the asymptotic ratio

$$\frac{c_{k+1}}{c_k} \approx \frac{1}{k}$$

This is the characteristic coefficient ratio of the Taylor series expansion for e^{ρ}. If the series were infinite, $v(\rho)$ would behave as e^{ρ} at large distances, causing the radial wave function $R(\rho) \sim e^{-\rho/2}e^{\rho} = e^{\rho/2}$ to diverge as $\rho \to \infty$.

To maintain a physically acceptable, normalizable wave function, the power series must terminate after a finite number of terms N, turning $v(\rho)$ into a polynomial of degree N. For the series to terminate at $k = N$, we must have $c_{N+1} = 0$, which requires

$$N + l + 1 - \rho_0 = 0$$

We define the principal quantum number n as

$$n = N + l + 1$$

where n takes positive integer values $n = 1, 2, 3, \ldots$, and for a given n, the orbital quantum number l can take integer values $l = 0, 1, 2, \ldots, n-1$.

This termination condition quantizes the parameter ρ_0:

$$\rho_0 = n$$

Recalling the definition of ρ_0, we equate it to n:

$$\frac{me^2}{4\pi\varepsilon_0\hbar^2\gamma} = n$$

Solving for γ gives

$$\gamma = \frac{me^2}{4\pi\varepsilon_0\hbar^2 n}$$

Defining the Bohr radius a_0 as

$$a_0 = \frac{4\pi\varepsilon_0\hbar^2}{me^2}$$

we write γ compactly as

$$\gamma = \frac{2}{na_0}$$

Substituting this expression into the variable transformation $\rho = \gamma r$ gives the dimensionless radial variable:

$$\rho = \frac{2r}{na_0}$$

We now substitute $\gamma = \frac{2}{na_0}$ back into the definition $\gamma^2 = -\frac{8mE}{\hbar^2}$:

$$\left(\frac{2}{na_0}\right)^2 = -\frac{8mE_n}{\hbar^2}$$

$$\frac{4}{n^2a_0^2} = -\frac{8mE_n}{\hbar^2}$$

Solving for the energy eigenvalues E_n yields

$$E_n = -\frac{\hbar^2}{2mn^2a_0^2}$$

Substituting the explicit formula for a_0 into the expression for E_n gives

$$E_n = -\frac{\hbar^2}{2mn^2}\left(\frac{me^2}{4\pi\varepsilon_0\hbar^2}\right)^2$$

$$E_n = -\frac{me^4}{2(4\pi\varepsilon_0)^2\hbar^2 n^2}, \quad n = 1, 2, 3, \ldots$$

Inserting the accepted values of the fundamental physical constants into this formula yields the quantized energy levels of the hydrogen atom:

$$E_n = -\frac{13.6}{n^2}\ \text{eV}$$

Complete Normalized Eigenfunctions

The terminated polynomial solutions $v(\rho)$ are related to the associated Laguerre polynomials $L_{n-l-1}^{2l+1}(\rho)$. Incorporating the normalization factor, the normalized radial wave functions $R_{nl}(r)$ are given by

$$R_{nl}(r) = \sqrt{\left(\frac{2}{na_0}\right)^3 \frac{(n-l-1)!}{2n(n+l)!}} e^{-\rho/2} \rho^l L_{n-l-1}^{2l+1}(\rho)$$

The complete normalized spatial eigenfunctions $\psi_{nlm_l}(r,\theta,\phi)$ of the hydrogen atom are formed by multiplying the radial functions by the spherical harmonics:

$$\psi_{nlm_l}(r,\theta,\phi) = R_{nl}(r) Y_l^{m_l}(\theta,\phi)$$

Explicit Radial Wave Functions R_{10}, R_{20}, and R_{21}

Evaluating the general formula for the lowest-lying electronic states yields the explicit radial functions:

1. For the ground state $n=1, l=0$ (1s state), we have $\rho = \frac{2r}{a_0}$ and $L_0^1(\rho) = 1$:

$$R_{10}(r) = 2\left(\frac{1}{a_0}\right)^{3/2} e^{-r/a_0}$$

1. For the excited state $n=2, l=0$ (2s state), we have $\rho = \frac{r}{a_0}$ and $L_1^1(\rho) = 2 - \frac{r}{a_0}$:

$$R_{20}(r) = \frac{1}{2\sqrt{2}}\left(\frac{1}{a_0}\right)^{3/2}\left(2 - \frac{r}{a_0}\right) e^{-r/(2a_0)}$$

1. For the excited state $n=2, l=1$ (2p state), we have $\rho = \frac{r}{a_0}$ and $L_0^3(\rho) = 1$:

$$R_{21}(r) = \frac{1}{2\sqrt{6}}\left(\frac{1}{a_0}\right)^{3/2}\left(\frac{r}{a_0}\right) e^{-r/(2a_0)}$$

3. Ground State of the Hydrogen Atom and Most Probable Position

The ground state of the hydrogen atom corresponds to the lowest energy state, characterized by the principal quantum number $n = 1$, orbital angular momentum quantum number $l = 0$, and magnetic quantum number $m_l = 0$. In this state, commonly referred to as the 1s state, the wave function has no angular dependence and is purely spherically symmetric. The normalized ground-state wave function $\psi_{100}(r,\theta,\phi)$ of the hydrogen atom is given by

$$\psi_{100}(r,\theta,\phi) = \frac{1}{\sqrt{\pi a_0^3}} e^{-r/a_0}$$

where r is the radial distance from the nucleus and a_0 is the Bohr radius, defined as

$$a_0 = \frac{4\pi\epsilon_0\hbar^2}{\mu e^2}$$

here μ is the reduced mass of the electron-proton system, e is the elementary charge, ϵ_0 is the permittivity of free space, and $\hbar$ is the reduced Planck constant.

To determine the spatial probability of finding the electron at a distance r from the nucleus, we consider a spherical shell of radius r and thickness dr. The volume element of this shell in spherical polar coordinates is given by $dV = 4\pi r^2 dr$. The radial probability density $P(r)$, defined such that $P(r)\,dr$ represents the probability of finding the electron between distance r and $r + dr$, is obtained by integrating the absolute square of the wave function over all angles θ and ϕ:

$$P(r)\,dr = \int_0^{2\pi}\int_0^{\pi} |\psi_{100}(r,\theta,\phi)|^2 r^2 \sin\theta\, d\theta\, d\phi\, dr$$

Since ψ_{100} is independent of θ and ϕ, the angular integration simply evaluates to 4π:

$$P(r)\,dr = 4\pi r^2 |\psi_{100}|^2\, dr$$

Substituting the explicit form of the ground-state wave function into the radial probability density expression gives

$$P(r)\,dr = 4\pi r^2 \left(\frac{1}{\sqrt{\pi a_0^3}} e^{-r/a_0}\right)^2 dr$$

Squaring the wave function inside the bracket yields

$$P(r)\,dr = 4\pi r^2 \left(\frac{1}{\pi a_0^3} e^{-2r/a_0}\right) dr$$

Canceling the factor of π in the numerator and denominator gives the simplified expression for the radial probability density function:

$$P(r) = \frac{4r^2}{a_0^3} e^{-2r/a_0}$$

To find the most probable position of the electron in the ground state, we maximize the radial probability density function $P(r)$ with respect to r. We set the first derivative of $P(r)$ with respect to r equal to zero:

$$\frac{dP(r)}{dr} = 0$$

Taking the derivative of $P(r)$ with respect to r using the product rule gives

$$\frac{dP(r)}{dr} = \frac{d}{dr}\left[\frac{4r^2}{a_0^3} e^{-2r/a_0}\right]$$

Factoring out the constant factor yields

$$\frac{dP(r)}{dr} = \frac{4}{a_0^3}\left[\frac{d}{dr}(r^2)e^{-2r/a_0} + r^2\frac{d}{dr}\left(e^{-2r/a_0}\right)\right]$$

Evaluating the individual derivatives inside the bracket yields

$$\frac{dP(r)}{dr} = \frac{4}{a_0^3}\left[2re^{-2r/a_0} + r^2\left(-\frac{2}{a_0}e^{-2r/a_0}\right)\right]$$

Factoring out $2re^{-2r/a_0}$ from both terms inside the bracket gives

$$\frac{dP(r)}{dr} = \frac{8r}{a_0^3}e^{-2r/a_0}\left(1 - \frac{r}{a_0}\right)$$

Setting this derivative equal to zero to determine the extrema:

$$\frac{8r}{a_0^3}e^{-2r/a_0}\left(1 - \frac{r}{a_0}\right) = 0$$

Since $r \neq 0$ and the exponential term $e^{-2r/a_0} \neq 0$ for finite r, the expression in the parenthesis must equal zero:

$$1 - \frac{r}{a_0} = 0$$

Solving for r gives the position of maximum radial probability:

$$r = a_0$$

Thus, the most probable distance of the electron from the nucleus in the ground state of the hydrogen atom is precisely equal to the Bohr radius a_0.

This result can be contrasted with the expectation value or average distance $\langle r \rangle$ of the electron from the nucleus in the ground state. By definition, the expectation value of r is calculated as

$$\langle r \rangle = \int_0^\infty rP(r)\,dr$$

Substituting the expression for $P(r)$ into the integral yields

$$\langle r \rangle = \int_0^\infty r\left(\frac{4r^2}{a_0^3}e^{-2r/a_0}\right)dr$$

Combining the powers of r inside the integral gives

$$\langle r \rangle = \frac{4}{a_0^3}\int_0^\infty r^3e^{-2r/a_0}\,dr$$

We change the variable of integration by introducing $x = \frac{2r}{a_0}$, which implies $r = \frac{a_0 x}{2}$ and $dr = \frac{a_0}{2}dx$. Substituting these into the integral gives

$$\langle r \rangle = \frac{4}{a_0^3} \int_0^\infty \left(\frac{a_0 x}{2}\right)^3 e^{-x} \left(\frac{a_0}{2} dx\right)$$

Extracting powers of a_0 and numerical constants from the integrand yields

$$\langle r \rangle = \frac{4}{a_0^3} \left(\frac{a_0^4}{16}\right) \int_0^\infty x^3 e^{-x}\, dx$$

Simplifying the constant factor outside the integral gives

$$\langle r \rangle = \frac{a_0}{4} \int_0^\infty x^3 e^{-x}\, dx$$

Using the standard gamma integral identity $\int_0^\infty x^n e^{-x}\, dx = n!$, we evaluate the integral for $n = 3$:

$$\int_0^\infty x^3 e^{-x}\, dx = 3! = 6$$

Substituting this value back into the expectation value equation yields

$$\langle r \rangle = \frac{a_0}{4} \times 6$$

Simplifying the fraction gives the final result for the average distance:

$$\langle r \rangle = \frac{3a_0}{2}$$

The average distance $\langle r \rangle = 1.5a_0$ is significantly larger than the most probable distance $r = a_0$. This difference arises because the radial probability density curve is asymmetric, featuring a long tail extending toward larger values of r, which shifts the mean value to a larger distance than the peak position.

Radial Probability Distribution Curves for 1s and 2s States

The radial probability distribution curve for the 1s state ($n = 1, l = 0, m_l = 0$) begins at $P(r) = 0$ at the origin $r = 0$, rises smoothly to a single distinct peak at $r = a_0$, and then decays exponentially toward zero as $r \to \infty$. The electron in the 1s state has no radial nodes, meaning there is no non-zero finite radius at which the radial wave function vanishes.

In contrast, the 2s state ($n = 2, l = 0, m_l = 0$) possesses a radial probability distribution curve characterized by two distinct peaks separated by a single radial node. The wave function for the 2s state vanishes at $r = 2a_0$, creating a nodal point where $P(r) = 0$. The inner peak of the 2s radial probability distribution occurs close to the nucleus (at $r \approx 0.76a_0$) and has a smaller amplitude. The outer peak is broader and reaches its maximum at a much larger distance ($r \approx 5.23a_0$). This structural difference reflects the higher principal quantum number n and the presence of $(n - l - 1) = (2 - 0 - 1) = 1$ radial node in the 2s state.

4. Degeneracy of the Energy Levels of the Hydrogen Atom

In the quantum mechanical treatment of the hydrogen atom, the energy eigenvalues E_n obtained by solving the Schrödinger equation depend exclusively on the principal quantum number n. The energy of a bound state with principal quantum number n is given by

$$E_n = -\frac{\mu e^4}{32\pi^2\epsilon_0^2\hbar^2 n^2} = -\frac{13.6 \text{ eV}}{n^2}$$

where $n = 1, 2, 3, \ldots$.

Although the energy eigenvalues depend only on n, a quantum state of the hydrogen atom is fully specified by three quantum numbers: the principal quantum number n, the orbital angular momentum quantum number l, and the magnetic quantum number m_l. For a given value of n, the allowed values of the orbital angular momentum quantum number l are

$$l = 0, 1, 2, \ldots, n-1$$

Furthermore, for each allowed value of l, the magnetic quantum number m_l can take $(2l+1)$ possible integer values ranging from $-l$ to $+l$:

$$m_l = -l, -(l-1), \ldots, 0, \ldots, l-1, l$$

Since the energy E_n does not depend on l or m_l, multiple distinct quantum states described by different pairs of (l, m_l) share the exact same energy eigenvalue E_n. The total number of linearly independent states associated with a single energy level E_n is called the degeneracy g_n of that energy level.

To determine the total degeneracy g_n for a given energy level n, we sum the number of allowed m_l states over all possible values of l from 0 to $n-1$:

$$g_n = \sum_{l=0}^{n-1} (2l+1)$$

We split this sum into two separate summations:

$$g_n = 2\sum_{l=0}^{n-1} l + \sum_{l=0}^{n-1} 1$$

The first term involves the sum of the first $(n-1)$ non-negative integers:

$$\sum_{l=0}^{n-1} l = \frac{(n-1)n}{2}$$

The second term represents adding the constant 1 a total of n times:

$$\sum_{l=0}^{n-1} 1 = n$$

Substituting these two evaluated terms back into the expression for g_n gives

$$g_n = 2\left[\frac{(n-1)n}{2}\right] + n$$

Canceling the factor of 2 in the first term yields

$$g_n = n(n-1) + n$$

Expanding the bracketed term gives

$$g_n = n^2 - n + n$$

Simplifying the expression results in the total degeneracy formula:

$$g_n = n^2$$

Thus, the energy level E_n of the hydrogen atom exhibits an n^2-fold degeneracy (excluding electron spin).

Physical Origin of the Degeneracy

The n^2-fold degeneracy of the hydrogen atom energy levels can be classified into two distinct physical origins:

1. **Geometric Degeneracy ($(2l+1)$-fold):** The independence of energy on the magnetic quantum number m_l is a direct consequence of the spherical symmetry (rotational invariance) of the central Coulomb potential $V(r) \propto 1/r$. Because the potential depends only on the radial distance r, space is isotropic, and the energy does not depend on the spatial orientation of the orbital angular momentum vector.
2. **Accidental Degeneracy:** The independence of energy on the orbital angular momentum quantum number l for a fixed principal quantum number n is unique to the exact $1/r$ dependence of the Coulomb potential (and the inverse-square force law). This additional symmetry, often termed "accidental degeneracy," is associated with a conserved vector specific to inverse-square central forces, known as the Laplace-Runge-Lenz vector. For central potentials other than $1/r$ (such as a spherical finite well), the energy levels depend on both n and l, breaking this additional degeneracy.

Degeneracy Values for Specific Energy Levels

We can evaluate the explicit state counts for the first three principal quantum levels:

For the ground state ($n = 1$): - Allowed orbital angular momentum value: $l = 0$ (s state). - Allowed magnetic quantum value: $m_l = 0$ (1 state). - Total number of states:

$$g_1 = 1^2 = 1 \text{ state}$$

The ground state is non-degenerate.

For the first excited state ($n = 2$): - Allowed orbital angular momentum values: $l = 0$ (2s) and $l = 1$ (2p). - For $l = 0$: $m_l = 0$ (1 state). - For $l = 1$: $m_l = -1, 0, +1$ (3 states). - Total number of states:

$$g_2 = 1 + 3 = 4 \text{ states}$$

This matches $g_2 = 2^2 = 4$.

For the second excited state ($n = 3$): - Allowed orbital angular momentum values: $l = 0$ (3s), $l = 1$ (3p), and $l = 2$ (3d). - For $l = 0$: $m_l = 0$ (1 state). - For $l = 1$: $m_l = -1, 0, +1$ (3 states). - For $l = 2$: $m_l = -2, -1, 0, +1, +2$ (5 states). - Total number of states:

$$g_3 = 1 + 3 + 5 = 9 \text{ states}$$

This matches $g_3 = 3^2 = 9$.

5. Linear Harmonic Oscillator

Time-Independent Schrödinger Equation for Harmonic Oscillator

A particle of mass m undergoing one-dimensional simple harmonic motion under a restoring force proportional to displacement experiences a potential energy given by

$$V(x) = \frac{1}{2} m \omega^2 x^2$$

where ω is the classical angular frequency of the oscillator. The time-independent Schrödinger equation for the particle in one dimension is

$$-\frac{\hbar^2}{2m} \frac{d^2\psi(x)}{dx^2} + V(x)\psi(x) = E\psi(x)$$

Substituting the expression for the potential energy $V(x)$ into the Schrödinger equation yields

$$-\frac{\hbar^2}{2m} \frac{d^2\psi(x)}{dx^2} + \frac{1}{2} m \omega^2 x^2 \psi(x) = E\psi(x)$$

Multiplying both sides of the equation by $-\frac{2m}{\hbar^2}$ gives

$$\frac{d^2\psi(x)}{dx^2} - \frac{m^2\omega^2}{\hbar^2} x^2 \psi(x) + \frac{2mE}{\hbar^2}\psi(x) = 0$$

To simplify this differential equation, we introduce a dimensionless independent variable ξ defined as

$$\xi = \sqrt{\frac{m\omega}{\hbar}} x$$

Differentiating ξ with respect to x gives

$$\frac{d\xi}{dx} = \sqrt{\frac{m\omega}{\hbar}}$$

Using the chain rule of differentiation, the first derivative of ψ with respect to x is expressed as

$$\frac{d\psi}{dx} = \frac{d\psi}{d\xi}\frac{d\xi}{dx} = \sqrt{\frac{m\omega}{\hbar}}\frac{d\psi}{d\xi}$$

Differentiating a second time with respect to x, we obtain

$$\frac{d^2\psi}{dx^2} = \frac{d}{dx}\left(\sqrt{\frac{m\omega}{\hbar}}\frac{d\psi}{d\xi}\right) = \sqrt{\frac{m\omega}{\hbar}}\frac{d^2\psi}{d\xi^2}\frac{d\xi}{dx} = \frac{m\omega}{\hbar}\frac{d^2\psi}{d\xi^2}$$

We also introduce a dimensionless energy parameter λ defined as

$$\lambda = \frac{2E}{\hbar\omega}$$

Substituting $\frac{d^2\psi}{dx^2} = \frac{m\omega}{\hbar}\frac{d^2\psi}{d\xi^2}$, $x^2 = \frac{\hbar}{m\omega}\xi^2$, and $\frac{2mE}{\hbar^2} = \frac{m\omega}{\hbar}\lambda$ into the differential equation gives

$$\frac{m\omega}{\hbar}\frac{d^2\psi}{d\xi^2} - \frac{m^2\omega^2}{\hbar^2}\left(\frac{\hbar}{m\omega}\xi^2\right)\psi + \frac{m\omega}{\hbar}\lambda\psi = 0$$

Dividing the entire equation by $\frac{m\omega}{\hbar}$ leads to the simplified differential equation

$$\frac{d^2\psi}{d\xi^2} + (\lambda - \xi^2)\psi = 0$$

Asymptotic Behavior and Solution Form

For very large values of ξ such that $\xi \to \pm\infty$, the parameter λ becomes negligible compared to ξ^2. In this asymptotic limit, the differential equation simplifies to

$$\frac{d^2\psi}{d\xi^2} - \xi^2\psi \approx 0$$

The approximate solutions to this asymptotic differential equation are of the form

$$\psi(\xi) \sim e^{\pm\xi^2/2}$$

To verify this, we differentiate $e^{-\xi^2/2}$ with respect to ξ:

$$\frac{d}{d\xi}\left(e^{-\xi^2/2}\right) = -\xi e^{-\xi^2/2}$$

Differentiating once more with respect to ξ yields

$$\frac{d^2}{d\xi^2}\left(e^{-\xi^2/2}\right) = \frac{d}{d\xi}\left(-\xi e^{-\xi^2/2}\right) = -e^{-\xi^2/2} + \xi^2 e^{-\xi^2/2} = (\xi^2 - 1)e^{-\xi^2/2}$$

For large values of ξ, $\xi^2 - 1 \approx \xi^2$, which confirms that $e^{-\xi^2/2}$ satisfies the asymptotic equation. Since physically acceptable wave functions must remain bounded as $\xi \to \pm\infty$, the solution $e^{+\xi^2/2}$ must be rejected as it diverges at infinity. Therefore, the physically acceptable asymptotic behavior is governed by $e^{-\xi^2/2}$.

To find an exact solution valid for all ξ, we express the wave function as the product of the asymptotic factor $e^{-\xi^2/2}$ and an unknown function $H(\xi)$:

$$\psi(\xi) = e^{-\xi^2/2} H(\xi)$$

Differentiating $\psi(\xi)$ with respect to ξ using the product rule gives

$$\frac{d\psi}{d\xi} = \frac{d}{d\xi}\left(e^{-\xi^2/2}H(\xi)\right) = -\xi e^{-\xi^2/2}H(\xi) + e^{-\xi^2/2}\frac{dH}{d\xi} = e^{-\xi^2/2}\left(\frac{dH}{d\xi} - \xi H\right)$$

Differentiating again to find the second derivative yields

$$\frac{d^2\psi}{d\xi^2} = \frac{d}{d\xi}\left[e^{-\xi^2/2}\left(\frac{dH}{d\xi} - \xi H\right)\right]$$

Expanding this derivative using the product rule gives

$$\frac{d^2\psi}{d\xi^2} = -\xi e^{-\xi^2/2}\left(\frac{dH}{d\xi} - \xi H\right) + e^{-\xi^2/2}\left(\frac{d^2H}{d\xi^2} - H - \xi\frac{dH}{d\xi}\right)$$

Factoring out $e^{-\xi^2/2}$ and combining like terms yields

$$\frac{d^2\psi}{d\xi^2} = e^{-\xi^2/2}\left(\frac{d^2H}{d\xi^2} - 2\xi\frac{dH}{d\xi} + (\xi^2 - 1)H\right)$$

Substituting $\psi(\xi) = e^{-\xi^2/2}H(\xi)$ and its second derivative into $\frac{d^2\psi}{d\xi^2} + (\lambda - \xi^2)\psi = 0$ gives

$$e^{-\xi^2/2}\left(\frac{d^2H}{d\xi^2} - 2\xi\frac{dH}{d\xi} + (\xi^2 - 1)H\right) + (\lambda - \xi^2)e^{-\xi^2/2}H = 0$$

Since $e^{-\xi^2/2}$ is never zero, we divide the equation by $e^{-\xi^2/2}$ to obtain

$$\frac{d^2H}{d\xi^2} - 2\xi\frac{dH}{d\xi} + (\xi^2 - 1)H + (\lambda - \xi^2)H = 0$$

Cancelling the $+\xi^2 H$ and $-\xi^2 H$ terms yields Hermite's differential equation:

$$\frac{d^2H}{d\xi^2} - 2\xi\frac{dH}{d\xi} + (\lambda - 1)H = 0$$

Power-Series Solution and Recurrence Relation

We solve Hermite's differential equation by assuming a power-series expansion for $H(\xi)$:

$$H(\xi) = \sum_{k=0}^{\infty} a_k \xi^k$$

Differentiating this series with respect to ξ gives

$$\frac{dH}{d\xi} = \sum_{k=1}^{\infty} k a_k \xi^{k-1}$$

Differentiating a second time yields

$$\frac{d^2H}{d\xi^2} = \sum_{k=2}^{\infty} k(k-1) a_k \xi^{k-2}$$

Shifting the summation index by replacing k with $k+2$ allows the second derivative to be written as

$$\frac{d^2H}{d\xi^2} = \sum_{k=0}^{\infty} (k+2)(k+1) a_{k+2} \xi^{k}$$

The middle term of Hermite's equation becomes

$$-2\xi \frac{dH}{d\xi} = -2\xi \sum_{k=1}^{\infty} k a_k \xi^{k-1} = \sum_{k=0}^{\infty} -2k a_k \xi^{k}$$

Substituting these series expansions into Hermite's differential equation gives

$$\sum_{k=0}^{\infty} (k+2)(k+1) a_{k+2} \xi^{k} - \sum_{k=0}^{\infty} 2k a_k \xi^{k} + \sum_{k=0}^{\infty} (\lambda - 1) a_k \xi^{k} = 0$$

Combining all terms into a single sum over powers of ξ^k yields

$$\sum_{k=0}^{\infty} \left[(k+2)(k+1) a_{k+2} + (\lambda - 1 - 2k) a_k \right] \xi^{k} = 0$$

For this equation to hold for all ξ, the coefficient of each power ξ^k must vanish independently:

$$(k+2)(k+1) a_{k+2} + (\lambda - 1 - 2k) a_k = 0$$

Rearranging this gives the recurrence relation for the coefficients:

$$a_{k+2} = \frac{2k + 1 - \lambda}{(k+2)(k+1)} a_k$$

Quantization Condition and Hermite Polynomials

The recurrence relation determines all even coefficients ($a_2, a_4, \ldots$) in terms of a_0, and all odd coefficients ($a_3, a_5, \ldots$) in terms of a_1. For large values of k, the ratio of successive terms in the series behaves as

$$\frac{a_{k+2}}{a_k} \approx \frac{2k}{k^2} = \frac{2}{k}$$

This asymptotic behavior is identical to the series expansion of e^{ξ^2}. If the series does not terminate, $H(\xi)$ will behave like e^{ξ^2} for large ξ. Consequently, the wave function $\psi(\xi) = e^{-\xi^2/2} H(\xi)$ would behave like

$e^{-\xi^2/2}e^{\xi^2} = e^{\xi^2/2}$ at large ξ, which diverges as $\xi \to \pm\infty$ and violates normalizability.

To make the wave function physically acceptable, the power series for $H(\xi)$ must terminate after a finite number of terms, turning $H(\xi)$ into a polynomial of degree n. For the series to terminate at $k = n$, the coefficient a_{n+2} must vanish, requiring the numerator of the recurrence relation to equal zero for $k = n$:

$$2n + 1 - \lambda = 0$$

This gives the quantization condition for λ:

$$\lambda = 2n + 1, \quad \text{where } n = 0, 1, 2, 3, \ldots$$

When $\lambda = 2n + 1$, the resulting polynomial solutions $H_n(\xi)$ of degree n are the Hermite polynomials. The Hermite polynomials are given by Rodrigues' formula:

$$H_n(\xi) = (-1)^n e^{\xi^2} \frac{d^n}{d\xi^n}\left(e^{-\xi^2}\right)$$

The first few Hermite polynomials are:

$$H_0(\xi) = 1$$

$$H_1(\xi) = 2\xi$$

$$H_2(\xi) = 4\xi^2 - 2$$

$$H_3(\xi) = 8\xi^3 - 12\xi$$

Energy Eigenvalues

Equating the quantization condition $\lambda = 2n + 1$ to the definition $\lambda = \frac{2E}{\hbar\omega}$ gives

$$\frac{2E}{\hbar\omega} = 2n + 1$$

Solving for the energy E yields the energy eigenvalues E_n:

$$E_n = \left(n + \frac{1}{2}\right)\hbar\omega, \quad \text{where } n = 0, 1, 2, 3, \ldots$$

The energy levels of the linear harmonic oscillator are discrete, non-degenerate, and equally spaced with a constant level separation of $\Delta E = E_{n+1} - E_n = \hbar\omega$.

Normalized Eigenfunctions

The unnormalized wave function corresponding to quantum number n is

$$\psi_n(\xi) = A_n e^{-\xi^2/2} H_n(\xi)$$

where A_n is the normalization constant. The normalization condition in coordinate space is

$$\int_{-\infty}^{\infty} |\psi_n(x)|^2 dx = 1$$

Converting the integral from x to ξ using $x = \sqrt{\frac{\hbar}{m\omega}}\xi$ and $dx = \sqrt{\frac{\hbar}{m\omega}}d\xi$ yields

$$|A_n|^2 \sqrt{\frac{\hbar}{m\omega}} \int_{-\infty}^{\infty} e^{-\xi^2} [H_n(\xi)]^2 d\xi = 1$$

Using the orthogonality integral for Hermite polynomials,

$$\int_{-\infty}^{\infty} e^{-\xi^2} H_n(\xi) H_m(\xi) d\xi = 2^n n! \sqrt{\pi}\, \delta_{nm}$$

for $m = n$, the integral evaluates to

$$\int_{-\infty}^{\infty} e^{-\xi^2} [H_n(\xi)]^2 d\xi = 2^n n! \sqrt{\pi}$$

Substituting this result into the normalization condition gives

$$|A_n|^2 \sqrt{\frac{\hbar}{m\omega}}\, 2^n n! \sqrt{\pi} = 1$$

Solving for $|A_n|^2$ yields

$$|A_n|^2 = \frac{1}{2^n n! \sqrt{\pi}} \sqrt{\frac{m\omega}{\hbar}}$$

Taking the positive square root gives the normalization constant

$$A_n = \frac{1}{\sqrt{2^n n!}} \left(\frac{m\omega}{\pi\hbar}\right)^{1/4}$$

Thus, the normalized eigenfunctions $\psi_n(x)$ of the linear harmonic oscillator are

$$\psi_n(x) = \frac{1}{\sqrt{2^n n!}} \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} e^{-\xi^2/2} H_n(\xi)$$

where $\xi = \sqrt{\frac{m\omega}{\hbar}}x$.

Ground State Wave Function and Zero-Point Energy

The ground state of the linear harmonic oscillator corresponds to the lowest quantum state, $n = 0$.

Substituting $n = 0$ into the energy eigenvalue formula $E_n = \left(n + \frac{1}{2}\right)\hbar\omega$ gives the ground state energy E_0:

$$E_0 = \left(0 + \frac{1}{2}\right)\hbar\omega = \frac{1}{2}\hbar\omega$$

This non-zero minimum energy $E_0 = \frac{1}{2}\hbar\omega$ is called the zero-point energy of the oscillator. It reflects the fundamental quantum mechanical feature that a trapped particle cannot have zero kinetic and potential energy simultaneously, in accordance with the Heisenberg uncertainty principle.

To find the ground state wave function $\psi_0(x)$, we substitute $n = 0$ and $H_0(\xi) = 1$ into the normalized eigenfunction expression:

$$\psi_0(x) = \frac{1}{\sqrt{2^0 0!}} \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} e^{-\xi^2/2}(1)$$

Since $2^0 = 1$ and $0! = 1$, this simplifies to

$$\psi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} e^{-\xi^2/2}$$

Substituting $\xi = \sqrt{\frac{m\omega}{\hbar}}x$ into the exponential factor gives the ground state wave function explicitly in terms of x:

$$\psi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} e^{-\frac{m\omega x^2}{2\hbar}}$$

The ground state wave function $\psi_0(x)$ is a symmetric Gaussian function centered at the origin $x = 0$, having no nodes, and decaying rapidly to zero as $x \to \pm\infty$.

Questions

Short form first (2–3 marks), then long form (5–10 marks). **There are 5 Assam University questions for this unit in the paper bank**, from CHMHCC-502T and CHMDSM-151, marked [AU] with their real source. The other questions are from other sources, marked as such.

Short-form questions

S1. [AU] Draw the radial probability distribution curves for 1s and 2s electrons. — CHMDSM-151, 2nd Sem, 2025, FYUG (Sec A).

Answer in brief: Section 3 — 1s: $P(r) = (4r^2/a_0^3)e^{-2r/a_0}$, a single peak at $r = a_0$; 2s: two peaks with a node at $r = 2a_0$.

S2. [non-AU] Verify that the average value of 1/r for a 1s electron in the hydrogen atom is 1/a0. — Univ. of Calicut, CBCSS B.Sc 5th Sem, PHY5 B07 Quantum Mechanics, Nov 2021, Q19 (5 marks).

Answer in brief: Section 3 — $\langle 1/r \rangle = \int_0^\infty (1/r)P(r)\,dr$ with $P(r) = (4r^2/a_0^3)e^{-2r/a_0}$ gives $1/a_0$.

S3. [non-AU] An electron is bound by a spring-like force (k = 89.5 eV/nm^2): (a) What is its ground-state energy? (b) How much energy to jump from the ground state to the second excited state? — Univ. of Calicut, CBCSS B.Sc 5th Sem, PHY5 B07 Quantum Mechanics, Nov 2021, Q15 (5 marks).

Answer in brief: Section 5 — $\omega = \sqrt{k/m}$, $E_0 = \frac{1}{2}\hbar\omega \approx 1.3$ eV; the jump to the second excited state ($n = 2$) needs $2\hbar\omega \approx 5.2$ eV.

S4. [standard] What is the degeneracy of the nth energy level of the hydrogen atom?

Answer in brief: Section 4 — n^2 states share the energy E_n: $l = 0, 1, \ldots, n-1$ with $m_l = -l, \ldots, +l$.

Long-form questions

L1. [AU] Write the Schrodinger wave equation in Cartesian coordinates (x,y,z), separate it in spherical polar coordinates (r,theta,phi) for the H-atom, and solve to find the most probable distance of the electron from the nucleus. — CHMHCC-502T, 5th Sem, 2020, CBCS (Sec B).

What to write: Sections 1 and 3 in full — the Cartesian equation with $V(r) = -e^2/(4\pi\epsilon_0 r)$, separation into radial and angular parts, $\psi_{100} = (1/\sqrt{\pi a_0^3})e^{-r/a_0}$, and $P(r)$ maximised at $r = a_0$.

L2. [AU] Write the Schrodinger equation for the H-atom, separate it in spherical polar coordinates to get the radial wave equation; and solve the Schrodinger equation for the one-dimensional harmonic oscillator (SHO) - what is its zero-point energy? — CHMHCC-502T, 5th Sem, 2021, CBCS (Sec B).

What to write: Sections 1 and 5 in full — the radial equation for $u(r) = rR(r)$; the oscillator solved by power series to $E_n = (n + \frac{1}{2})\hbar\omega$ with zero-point energy $E_0 = \frac{1}{2}\hbar\omega$.

L3. [AU] Write the Schrodinger equation for the hydrogen atom and separate it in spherical polar coordinates; solve for H and find the most probable distance of the electron from the nucleus. — CHMHCC-502T, 5th Sem, 2023, CBCS (Sec B).

What to write: Sections 1–3 in full — separation, the energy eigenvalues $E_n = -13.6/n^2$ eV, the ground-state wave function, and the most probable distance $r = a_0$.

L4. [standard] Derive the energy eigenvalues of the hydrogen atom using spherical harmonics and Laguerre polynomials, give the ground-state eigenfunction, and explain the degeneracy of the energy levels.

What to write: Sections 2 and 4 in full — the radial power series, associated Laguerre polynomials, $E_n = -13.6/n^2$ eV, $\psi_{nlm_l} = R_{nl}(r)Y_l^{m_l}(\theta, \phi)$, and the n^2-fold degeneracy.

L5. [standard] Derive the energy eigenvalues and eigenfunctions of the linear harmonic oscillator and write down its ground-state wave function.

What to write: Section 5 in full — power-series solution, Hermite polynomials, $E_n = (n + \frac{1}{2})\hbar\omega$, $\psi_n(x)$ with $H_n(\xi)$, and $\psi_0(x) = (m\omega/\pi\hbar)^{1/4}e^{-m\omega x^2/2\hbar}$.

Sources

These notes were written from the unit syllabus only, as the user asked. No external PDF, textbook, or lecture notes were used. The questions are from the Assam University paper bank and other universities' papers (see the Questions section above), not from any source used for the notes.